

Node Structure Visualization

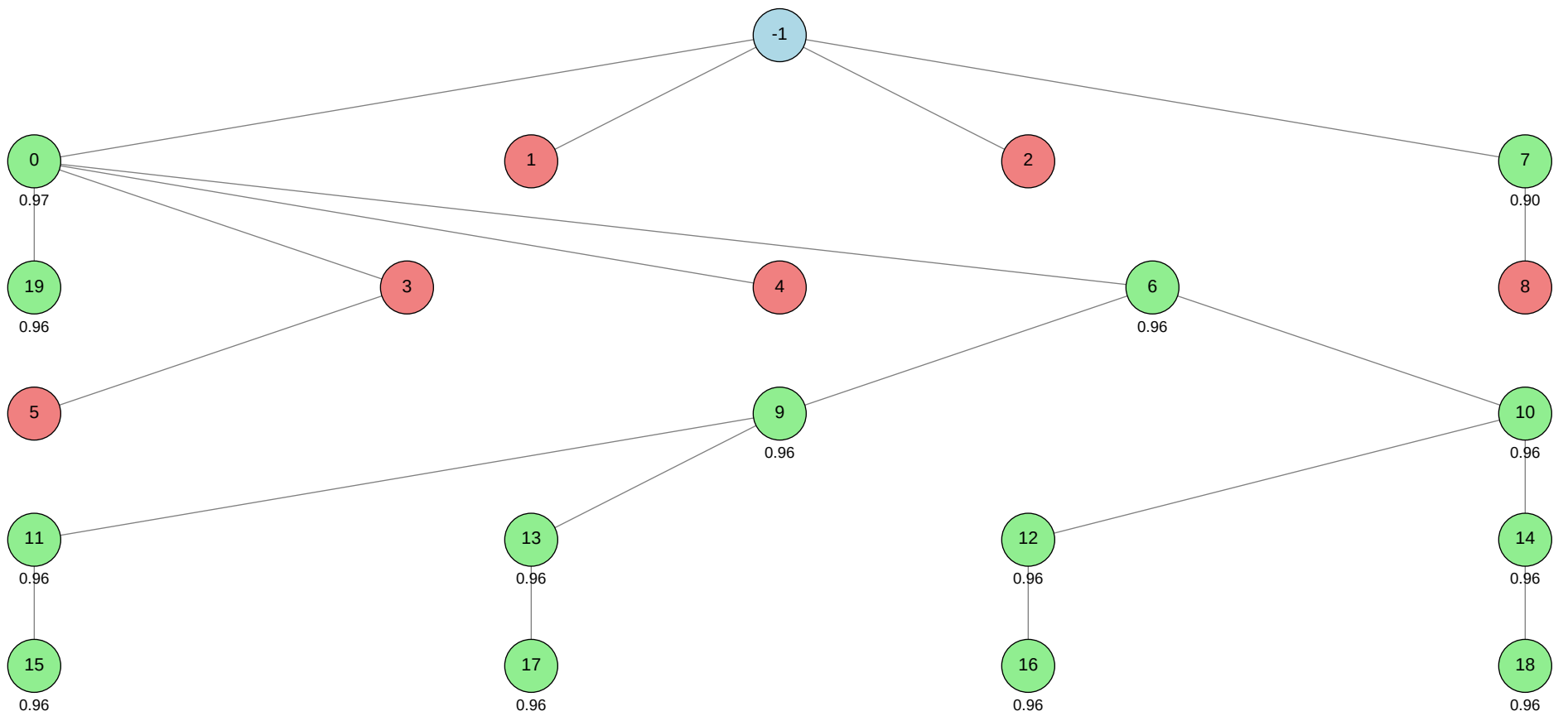
Click on any node in the tree to jump to its detailed information.

Total Nodes: 21 | Best Validation Score: 0.9654 (Node 0)

Node Tree:

Table of Contents:

Node ID	Stage	Status	Tool
-1			
0			autogluon.multimodal
1			machine learning
2			autogluon.tabular
3			autogluon.multimodal
4			autogluon.multimodal
5			autogluon.multimodal
6			autogluon.multimodal
7			machine learning
8			machine learning
9			autogluon.multimodal
10			autogluon.multimodal
11			autogluon.multimodal
12			autogluon.multimodal
13			autogluon.multimodal
14			autogluon.multimodal
15			autogluon.multimodal
16			autogluon.multimodal
17			autogluon.multimodal
18			autogluon.multimodal
19			autogluon.multimodal



Status	Color
Success	Green
Failure	Red
Neutral	Blue

Node Details:

Node -1 (root)

Property	Value
uct_value	0.5137
ctime	2026-05-21 16:14:25
debug_attempts	0
depth	0
execution_time	0.0
expected_child_count	0
failure_visits	6
id	-1
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✗
num_children	4
prev_tutorial_prompt	None
stage	root
time_step	-1
tool_used	
tools_available	[]
unvalidated_visits	0
validated_reward	13.407259999999999
validated_visits	14
validation_score	None
visits	20
parent_id	None
child_ids	[0, 1, 2, 7]

Node 0 (evolve)

Property	Value
uct_value	1.3477
avg_raw_score	0.9619383333333333
ctime	2026-05-21 16:15:22
debug_attempts	0
depth	1
execution_time	0.0
expected_child_count	0
exploitation	0.7222433661063543
exploration	0.6264768959394785
failure_penalty	-0.03333333333333333
failure_visits	3
id	0
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	1
normalized_score	0.9444708742996095
num_children	4
prev_tutorial_prompt	None
stage	evolve
time_step	0
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.7555766994396876
validated_reward	12.50488

validated_visits	13
validated_weight	0.8
validation_score	0.9654
visits	16
parent_id	-1
child_ids	[19, 3, 4, 6]

Node 1 (evolve)

Property	Value
uct_value	2.4474
ctime	2026-05-21 17:16:55
debug_attempts	0
depth	1
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.4793538436473155
failure_penalty	-0.0
failure_visits	1
id	1
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	0
prev_tutorial_prompt	None
stage	evolve
time_step	1
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	1
parent_id	-1
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

warning: Requirements file `~/home/anri21/.conda/envs/mlzero\_env/lib/python3.11/site-packages/autogluon/assistant/tools\_registry/machine learning/requirements.txt` does not contain any dependencies

Using Python 3.11.15 environment at: 45-addition_2/node_1/conda_env

Resolved 79 packages in 333ms

Uninstalled 1 package in 1.87s

Installed 78 packages in 37.28s

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiosignal==1.4.0

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ anyio==4.13.0

+ attrs==26.1.0

+ certifi==2026.5.20

+ chardet==7.4.3

+ charset-normalizer==3.4.7

+ click==8.4.0

+ cuda-bindings==13.2.0

+ cuda-pathfinder==1.5.4

+ cuda-toolkit==13.0.2

+ datasets==4.8.5

+ dill==0.4.1

+ filelock==3.29.0

+ frozenlist==1.8.0

+ fsspec==2026.2.0

+ h11==0.16.0

+ hf-xet==1.5.0

+ httpcore==1.0.9

+ httpx==0.28.1

+ huggingface-hub==1.16.0

+ idna==3.15

+ jinja2==3.1.6


```
+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocessing==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.13.4
+ pydantic-core==2.46.4
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2
+ rich==15.0.0
- setuptools==82.0.1
+ set... [truncated, 108861 chars total]
```

Error Analysis:

ERROR_SUMMARY: The main error is that the training process exceeded the 1-hour (3600 seconds) wall-clock time limit before completing all epochs, causing the script to be killed and preventing subsequent prediction, saving, and validation steps. Additionally, there are repeated `OSError: [Errno 16] Device or resource busy` errors related to deletion of `.nfs*` files (likely due to NFS file system locks), and warnings about excessive DataLoader worker processes (32 used vs. 2 recommended), which may cause slowness or instability.

SUGGESTED_FIX:

Reduce the number of DataLoader worker processes to 2 or fewer to avoid resource contention and potential slowness. Lower the batch size and/or the number of training epochs to ensure the entire script (including data loading, training, and inference) completes well within the 1-hour time limit. Also, avoid unnecessary file deletions or ensure files are not being accessed elsewhere to prevent .nfs* OSError issues.

Node 2 (evolve)

Property	Value
uct_value	2.4474
ctime	2026-05-21 18:19:45
debug_attempts	0
depth	1
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.2131672923786587
failure_penalty	-0.0
failure_visits	1
id	2
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✗
normalized_failure_visit	0
num_children	0
prev_tutorial_prompt	None
stage	evolve
time_step	2
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	1
parent_id	-1
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 45-addition_2/node_2/conda_env

Resolved 239 packages in 3.76s

Uninstalled 1 package in 1.77s

Installed 237 packages in 2m 05s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260521

+ autogluon-common==1.5.1b20260521

+ autogluon-core==1.5.1b20260521

+ autogluon-features==1.5.1b20260521

+ autogluon-multimodal==1.5.1b20260521

+ autogluon-tabular==1.5.1b20260521

+ autogluon-timeseries==1.5.1b20260521

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.12

+ botocore==1.43.12

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.5.20

+ cffi==2.0.0

- + chardet==7.4.3
- + charset-normalizer==3.4.7
- + chronos-forecasting==2.2.2
- + click==8.4.0
- + cloudpathlib==0.24.0
- + cloudpickle==3.1.2
- + colorama==0.4.6
- + colorful==0.6.0a1
- + colorlog==6.10.1
- + confection==1.3.3
- + contourpy==1.3.3
- + coreforecast==0.0.16
- + cryptography==48.0.0
- + cycler==0.12.1
- + cymem==2.0.13
- + daal==2025.9.0
- + datasets==4.0.0
- + defusedxml==0.7.1
- + dill==0.3.8
- + distlib==0.4.0
- + einops==0.8.2
- + einx==0.4.3
- + evaluate==0.4.6
- + fastai==2.8.5
- + fastcore==1.8.18
- + fastdownload==0.0.7
- + fastprogress==1.0.5
- + fasttransform==0.0.2
- + filelock==3.29.0
- + fonttools==4.63.0
- + frozendict==2.4.7
- + frozenlist==1.8.0
- + fsspec==2025.3.0
- + fugue==0.9.7
- + future==1.0.0
- + gdown==6.0.0
- + gluonts==0.16.2
- + google-api-core==2.30.3
- + google-auth==2.53.0
- + googleapis-common-protos==1.75.0
- + graphviz==0.21
- + grpcio==1.81.0rc1
- + hf-xet==1.5.0
- + httpx==1.0.dev3
- + huggingface-hub==0.36.2

+ hyperopt==0.... [truncated, 41210 chars total]

Error Analysis:

ERROR_SUMMARY: The root cause is that AutoGluon Tabular attempted to use Ray for parallel model training, but the UNIX socket path for Ray's plasma_store exceeded the system's 107-character limit, causing OSError and resulting in all major models failing to train. Only a fallback model (Mitra_c1_BAG_L1) was used, which likely produced no predictions, leading to a downstream ValueError ("Empty data passed with indices specified") during prediction.

SUGGESTED_FIX:

Shorten the directory paths used for temporary files and model output to ensure all UNIX socket paths are well under 107 characters (e.g., move your working/output directories to a higher-level folder with a shorter absolute path). Alternatively, set the environment variable `TMPDIR` or Ray's `temp\_dir` to a short path (like `/tmp/agtmp`) before running your script, and ensure all AutoGluon/Ray temp files are created there. After adjusting paths, rerun the training script to allow Ray and AutoGluon to create valid sockets and successfully train all models.

Node 3 (evolve)

Property	Value
uct_value	1.6649
ctime	2026-05-21 18:31:20
debug_attempts	1
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.6453669408194262
failure_penalty	-0.0
failure_visits	2
id	3
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	1
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	3
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0

validated_visits	0
validation_score	None
visits	2
parent_id	0
child_ids	[5]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 45-addition_2/node_3/conda_env

Resolved 235 packages in 3.37s

Prepared 1 package in 340ms

Uninstalled 1 package in 1.95s

Installed 233 packages in 2m 00s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260521

+ autogluon-common==1.5.1b20260521

+ autogluon-core==1.5.1b20260521

+ autogluon-features==1.5.1b20260521

+ autogluon-multimodal==1.5.1b20260521

+ autogluon-tabular==1.5.1b20260521

+ autogluon-timeseries==1.5.1b20260521

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.12

+ botocore==1.43.12

- + catalogue==2.0.10
- + catboost==1.2.10
- + certifi==2026.5.20
- + cffi==2.0.0
- + chardet==7.4.3
- + charset-normalizer==3.4.7
- + chronos-forecasting==2.2.2
- + click==8.4.0
- + cloudpathlib==0.24.0
- + cloudpickle==3.1.2
- + colorama==0.4.6
- + colorful==0.6.0a1
- + colorlog==6.10.1
- + confection==1.3.3
- + contourpy==1.3.3
- + coreforecast==0.0.16
- + cryptography==48.0.0
- + cycler==0.12.1
- + cymem==2.0.13
- + datasets==4.0.0
- + defusedxml==0.7.1
- + dill==0.3.8
- + distlib==0.4.0
- + einops==0.8.2
- + einx==0.4.3
- + evaluate==0.4.6
- + fastai==2.8.5
- + fastcore==1.8.18
- + fastdownload==0.0.7
- + fastprogress==1.0.5
- + fasttransform==0.0.2
- + filelock==3.29.0
- + fonttools==4.63.0
- + frozendict==2.4.7
- + frozenlist==1.8.0
- + fsspec==2025.3.0
- + fugue==0.9.7
- + future==1.0.0
- + gdown==6.0.0
- + gluonts==0.16.2
- + google-api-core==2.30.3
- + google-auth==2.53.0
- + googleapis-common-protos==1.75.0
- + graphviz==0.21
- + grpcio==1.81.0rc1

+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hy... [truncated, 18812 chars total]

Error Analysis:

ERROR_SUMMARY: The root cause of the failure is a distributed training error: ``torch.distributed.DistNetworkError`` ("address already in use" on port 20619) during Distributed Data Parallel (DDP) initialization in PyTorch, which prevented the model from starting training. This is likely due to the specified port already being occupied by another process or a previous run that did not clean up properly. Additional secondary errors include resource cleanup warnings (leaked semaphores, NFS file busy errors) and a ``pickle.UnpicklingError`` during multiprocessing shutdown, which are likely consequences of the main distributed initialization failure.

SUGGESTED_FIX:

Change the DDP initialization port to a random or unused value by setting the ``MASTER_PORT`` environment variable before training, or configure AutoGluon/PyTorch to use a different port for distributed training. Ensure that no other processes are occupying the port, and that any previous failed runs have fully released resources. If running multiple jobs on the same node, assign unique ports per job and consider reducing the number of GPUs/workers if port conflicts persist.

Node 4 (evolve)

Property	Value
uct_value	2.3545
ctime	2026-05-21 18:43:27
debug_attempts	0
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.664857771836881
failure_penalty	-0.0
failure_visits	1
id	4
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	4
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0

validated_visits	0
validation_score	None
visits	1
parent_id	0
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 45-addition_2/node_4/conda_env

Resolved 235 packages in 3.65s

Prepared 4 packages in 937ms

Uninstalled 1 package in 2.09s

Installed 233 packages in 1m 51s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260521

+ autogluon-common==1.5.1b20260521

+ autogluon-core==1.5.1b20260521

+ autogluon-features==1.5.1b20260521

+ autogluon-multimodal==1.5.1b20260521

+ autogluon-tabular==1.5.1b20260521

+ autogluon-timeseries==1.5.1b20260521

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.12

+ botocore==1.43.12

- + catalogue==2.0.10
- + catboost==1.2.10
- + certifi==2026.5.20
- + cffi==2.0.0
- + chardet==7.4.3
- + charset-normalizer==3.4.7
- + chronos-forecasting==2.2.2
- + click==8.4.0
- + cloudpathlib==0.24.0
- + cloudpickle==3.1.2
- + colorama==0.4.6
- + colorful==0.6.0a1
- + colorlog==6.10.1
- + confection==1.3.3
- + contourpy==1.3.3
- + coreforecast==0.0.16
- + cryptography==48.0.0
- + cycler==0.12.1
- + cymem==2.0.13
- + datasets==4.0.0
- + defusedxml==0.7.1
- + dill==0.3.8
- + distlib==0.4.0
- + einops==0.8.2
- + einx==0.4.3
- + evaluate==0.4.6
- + fastai==2.8.5
- + fastcore==1.8.18
- + fastdownload==0.0.7
- + fastprogress==1.0.5
- + fasttransform==0.0.2
- + filelock==3.29.0
- + fonttools==4.63.0
- + frozendict==2.4.7
- + frozenlist==1.8.0
- + fsspec==2025.3.0
- + fugue==0.9.7
- + future==1.0.0
- + gdown==6.0.0
- + gluonts==0.16.2
- + google-api-core==2.30.3
- + google-auth==2.53.0
- + googleapis-common-protos==1.75.0
- + graphviz==0.21
- + grpcio==1.81.0rc1

+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ h... [truncated, 18849 chars total]

Error Analysis:

ERROR_SUMMARY: The root cause of the error is that PyTorch's distributed training backend (`torch.distributed`) failed to initialize because the specified TCP port (20619) was already in use (`EADDRINUSE`). This prevented the distributed process group from starting, causing the entire training process to fail. Additionally, there are secondary errors related to resource cleanup (leaked semaphores, NFS file busy errors) and a `pickle` unpickling error, which are likely consequences of the failed distributed/multiprocessing setup.

SUGGESTED_FIX:

Change the distributed training configuration to avoid port conflicts: set `"env.strategy": "auto"` or `"env.strategy": "ddp"` with `"env.dist_port"` set to a random or unused port in the `hyperparameters` dictionary, or set `"env.num_gpus": 1` to disable distributed training if only one GPU is needed. Before rerunning, ensure no zombie processes are holding onto ports (kill any leftover Python processes), and consider reducing `"env.num_workers"` to minimize NFS and semaphore issues. If running multiple jobs on the same node, coordinate port assignments or use `"env.dist_port": 0` (for random port selection, if supported).

Node 5 (debug)

Property	Value
uct_value	1.1772
ctime	2026-05-21 18:54:30
debug_attempts	1
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	1
id	5
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	debug
time_step	5
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	3
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 45-addition_2/node_5/conda_env

Resolved 235 packages in 3.25s

Uninstalled 1 package in 2.03s

Installed 233 packages in 1m 44s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260521

+ autogluon-common==1.5.1b20260521

+ autogluon-core==1.5.1b20260521

+ autogluon-features==1.5.1b20260521

+ autogluon-multimodal==1.5.1b20260521

+ autogluon-tabular==1.5.1b20260521

+ autogluon-timeseries==1.5.1b20260521

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.12

+ botocore==1.43.12

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.5.20

+ cffi==2.0.0

+ chardet==7.4.3

+ charset-normalizer==3.4.7

+ chronos-forecasting==2.2.2

+ click==8.4.0

+ cloudpathlib==0.24.0

+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.2.7
+ idna==3.15
... [truncated, 17812 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurs because the test data passed to ``predictor.predict_proba`` only contains the ``image_name`` column, but the model was trained using additional tabular features (``skin_tone`` and ``alternative_skin_tone``). AutoGluon MultiModalPredictor requires all columns used during training to also be present in the test data at prediction time, resulting in a `ValueError` when these columns are missing.

SUGGESTED_FIX:

Modify your test data preparation so that the `DataFrame` passed to ``predictor.predict_proba`` includes the columns ``skin_tone`` and ``alternative_skin_tone`` (with appropriate values). If these features are not available for the test set, add them with placeholder values (e.g., the mode or a neutral value used during training) to ensure the test `DataFrame` matches the training feature schema. Double-check that the column order and names exactly match those used during training.

Node 6 (debug)

Property	Value
uct_value	1.6493
avg_raw_score	0.96162
ctime	2026-05-21 19:56:47
debug_attempts	1
depth	3
execution_time	0.0
expected_child_count	0
exploitation	0.9394227719632107
exploration	0.7015868209010818
failure_penalty	-0.0
failure_visits	0
id	6
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9394227719632107
num_children	2
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	debug
time_step	6
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9394227719632107
validated_reward	10.577820000000001
validated_visits	11
validated_weight	1.0
validation_score	0.9616
visits	11
parent_id	0
child_ids	[9, 10]

Node 7 (debug)

Property	Value
uct_value	1.7306
avg_raw_score	0.90238
ctime	2026-05-21 20:59:02
debug_attempts	1
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.2867582287320087
failure_penalty	-0.0
failure_visits	1
id	7
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.0
num_children	1
prev_tutorial_prompt	None
stage	debug
time_step	7
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.90238

validated_visits	1
validated_weight	0.5
validation_score	0.9024
visits	2
parent_id	-1
child_ids	[8]

Node 8 (evolve)

Property	Value
uct_value	1.1772
ctime	2026-05-21 21:56:16
debug_attempts	0
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	1
id	8
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have them
stage	evolve
time_step	8
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0

validation_score	None
visits	1
parent_id	7
child_ids	[]

Error Message:

```

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `~/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt`
does not contain any dependencies
Using Python 3.11.15 environment at: 45-addition_2/node_8/conda_env
Resolved 79 packages in 1.10s
Uninstalled 1 package in 1.65s
Installed 78 packages in 30.55s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.0
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0
+ frozenlist==1.8.0
+ fsspec==2026.2.0
+ h11==0.16.0
+ hf-xet==1.5.0
+ httpcore==1.0.9
+ httpx==0.28.1
+ huggingface-hub==1.16.1
+ idna==3.15

```



```
+ jinja2==3.1.6
+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocessing==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.13.4
+ pydantic-core==2.46.4
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2
+ rich==15.0.0
- setuptools==82.0.1
+ set... [truncated, 118694 chars total]
```

Error Analysis:

ERROR_SUMMARY: The process was terminated because it exceeded the 1-hour (3600 seconds) wall-clock time limit before completing model training, test prediction, and result saving. The main bottleneck is slow training, likely due to inefficient use of hardware (CPU instead of GPU, or slow data loading), a large dataset, and/or a model (EfficientNet-B0) and training loop that are too slow for the time constraints.

SUGGESTED_FIX:

Reduce the number of training epochs (e.g., from 10 to 2–3), use a smaller model (e.g., ResNet18 or MobileNetV2), and/or decrease the training set size (e.g., subsample the data or use a smaller validation split). Also, ensure that the script is actually using the GPU (check CUDA availability and driver compatibility), and consider increasing the batch size if memory allows to speed up training.

Node 9 (evolve)

Property	Value
uct_value	1.9186
avg_raw_score	0.9616200000000001
ctime	2026-05-21 22:58:44
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
exploitation	0.9394227719632124
exploration	0.9595602560122969
failure_penalty	-0.0
failure_visits	0
id	9
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9394227719632124
num_children	2
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	9
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9394227719632124
validated_reward	4.8081000000000005
validated_visits	5
validated_weight	1.0
validation_score	0.9616
visits	5
parent_id	6
child_ids	[11, 13]

Node 10 (evolve)

Property	Value
uct_value	1.9186
avg_raw_score	0.96162
ctime	2026-05-22 00:01:45
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
exploitation	0.9394227719632107
exploration	1.0728209804752984
failure_penalty	-0.0
failure_visits	0
id	10
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9394227719632107
num_children	2
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	10
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9394227719632107
validated_reward	4.8081000000000005
validated_visits	5
validated_weight	1.0
validation_score	0.9616
visits	5
parent_id	6
child_ids	[12, 14]

Node 11 (evolve)

Property	Value
uct_value	2.2079
avg_raw_score	0.96162
ctime	2026-05-22 01:03:42
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
exploitation	0.9394227719632107
exploration	1.1772322201769845
failure_penalty	-0.0
failure_visits	0
id	11
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9394227719632107
num_children	1
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	11
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9394227719632107
validated_reward	1.92324
validated_visits	2
validated_weight	1.0
validation_score	0.9616
visits	2
parent_id	9
child_ids	[15]

Node 12 (evolve)

Property	Value
uct_value	2.2079
avg_raw_score	0.96162
ctime	2026-05-22 02:05:23
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
exploitation	0.9394227719632107
exploration	1.1772322201769845
failure_penalty	-0.0
failure_visits	0
id	12
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9394227719632107
num_children	1
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	12
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9394227719632107
validated_reward	1.92324
validated_visits	2
validated_weight	1.0
validation_score	0.9616
visits	2
parent_id	10
child_ids	[16]

Node 13 (evolve)

Property	Value
uct_value	2.2079
avg_raw_score	0.96162
ctime	2026-05-22 03:07:15
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
exploitation	0.9394227719632107
exploration	1.664857771836881
failure_penalty	-0.0
failure_visits	0
id	13
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9394227719632107
num_children	1
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	13
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9394227719632107
validated_reward	1.92324
validated_visits	2
validated_weight	1.0
validation_score	0.9616
visits	2
parent_id	9
child_ids	[17]

Node 14 (evolve)

Property	Value
uct_value	2.2079
avg_raw_score	0.96162
ctime	2026-05-22 04:08:53
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
exploitation	0.9394227719632107
exploration	1.664857771836881
failure_penalty	-0.0
failure_visits	0
id	14
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9394227719632107
num_children	1
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	14
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9394227719632107
validated_reward	1.92324
validated_visits	2
validated_weight	1.0
validation_score	0.9616
visits	2
parent_id	10
child_ids	[18]

Node 15 (evolve)

Property	Value
uct_value	2.1167
ctime	2026-05-22 05:10:44
debug_attempts	0
depth	6
execution_time	0.0
expected_child_count	0
failure_visits	0
id	15
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	15
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.96162
validated_visits	1
validation_score	0.9616
visits	1
parent_id	11
child_ids	[]

Node 16 (evolve)

Property	Value
uct_value	2.1167
ctime	2026-05-22 06:13:01
debug_attempts	0
depth	6
execution_time	0.0
expected_child_count	0
failure_visits	0
id	16
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	16
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.96162
validated_visits	1
validation_score	0.9616
visits	1
parent_id	12
child_ids	[]

Node 17 (evolve)

Property	Value
uct_value	2.1167
ctime	2026-05-22 07:14:54
debug_attempts	0
depth	6
execution_time	0.0
expected_child_count	0
failure_visits	0
id	17
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	17
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.96162
validated_visits	1
validation_score	0.9616
visits	1
parent_id	13
child_ids	[]

Node 18 (evolve)

Property	Value
uct_value	2.1167
ctime	2026-05-22 08:17:58
debug_attempts	0
depth	6
execution_time	0.0
expected_child_count	0
failure_visits	0
id	18
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	18
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.96162
validated_visits	1
validation_score	0.9616
visits	1
parent_id	14
child_ids	[]

Node 19 (debug)

Property	Value
uct_value	3.2939
ctime	2026-05-22 09:20:25
debug_attempts	2
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	0
id	19
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	debug
time_step	19
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.96162
validated_visits	1
validation_score	0.9616
visits	1
parent_id	0
child_ids	[]

